Problem 1

a) Yes

Suppose $x, y \in S$, $x = x_1 + x_2$, $x_1 \in S_1$, $x_2 \in S_2$

$y = y_1 + y_2$, $y_1 \in S_1$, $y_2 \in S_2$

For any $\lambda \in [0,1]$, $\lambda x + (1-\lambda) y$

$= \lambda x_1 + (1-\lambda) y_1 + \lambda x_2 + (1-\lambda) y_2$

$= z_1 + z_2$, with $z_1 = \lambda x_1 + (1-\lambda) y_1$, $z_2 = \lambda x_2 + (1-\lambda) y_2$

$\because x_1, y_1 \in S_1$, S_1 is convex

$\therefore z_1 = \lambda x_1 + (1-\lambda) y_1 \in S_1$

$\because x_2, y_2 \in S_2$, S_2 is convex

$\therefore z_2 = \lambda x_2 + (1-\lambda) y_2 \in S_2$

$\therefore x + y = z_1 + z_2$, with $z_1 \in S_1$, $z_2 \in S_2$

$\therefore x + y \in S$

$\therefore$ We can prove that S is convex.

b) No.

Assuming that $x_1^2 + y_1^2 = r$, $x_2^2 + y_2^2 = r$, $x_1, x_2, y_1, y_2 \in \mathbb{R}$

$\therefore \forall \lambda \in [0,1]$, $[\lambda x_1 + (1-\lambda) x_2]^2 + [\lambda y_1 + (1-\lambda) y_2]^2$

$= \lambda^2 (x_1^2 + y_1^2) + (1-\lambda)^2 (x_2^2 + y_2^2) + 2\lambda(1-\lambda)(x_1 x_2 + y_1 y_2)$

$= r\lambda^2 + r(1-\lambda)^2 + 2\lambda(1-\lambda)(x_1 x_2 + y_1 y_2)$

$= r + 2\lambda(1-\lambda)(x_1 x_2 + y_1 y_2)$

$\therefore$ When $\lambda(1-\lambda)(x_1 x_2 + y_1 y_2) \neq 0$,

$\lambda \cdot (x_1, y_1) + (1-\lambda) \cdot (x_2, y_2) \notin \{(x,y) \in \mathbb{R}^2 : x^2 + y^2 = r\}$

$\therefore$ The set is not convex.

c) No. For $ax^2 + 2x - \frac{1}{a} \geq 0$

$\Delta = 2^2 + 4 \cdot a \cdot \frac{1}{a} = 4 + 4 = 8 > 0$

$\therefore a x^2 + 2x - \frac{1}{a} \geq 0 \Rightarrow x^2 + \frac{2}{a} x + \frac{1}{a^2} \geq \frac{2}{a^2} \Rightarrow (x + \frac{1}{a})^2 \geq \frac{2}{a^2}$

$\Rightarrow \quad x \in (-\infty, \frac{-\sqrt{2}-1}{a}] \cup [\frac{\sqrt{2}-1}{a}, \infty)$

A counter example:

$\because \frac{1}{2}(\frac{-\sqrt{2}-1}{a} + \frac{\sqrt{2}-1}{a}) = -\frac{1}{a} \notin (-\infty, \frac{-\sqrt{2}-1}{a}] \cup [\frac{\sqrt{2}-1}{a}, \infty)$

$\therefore (-\infty, \frac{-\sqrt{2}-1}{a}] \cup [\frac{\sqrt{2}-1}{a}, \infty)$ is not convex

$\therefore \{x \in \mathbb{R} : ax^2 + 2x - \frac{1}{a} \geq 0\}$ with $a > 0$ is not convex

d) Yes.

Assuming that $y_1 \geq |x_1|$, $y_2 \geq |x_2|$

$\therefore \forall \lambda \in [0,1]$, $\lambda(x_1, y_1) + (1-\lambda)(x_2, y_2)$

$= (\lambda x_1 + (1-\lambda) x_2, \lambda y_1 + (1-\lambda) y_2)$

$\because y_1 \geq |x_1|$, $y_2 \geq |x_2|$

$\therefore \lambda y_1 + (1-\lambda) y_2 \geq \lambda |x_1| + (1-\lambda) |x_2| \geq |\lambda x_1 + (1-\lambda) x_2|$

$\therefore \lambda(x_1, y_1) + (1-\lambda)(x_2, y_2) \in \{(x,y) \in \mathbb{R}^2 : y \geq |x|\}$

$\therefore$ The set is convex.

e) No

Suppose $x^T A x \leq 0$, $y^T A y \leq 0$

$x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$, $y = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$

$\therefore \forall \lambda \in [0,1]$, $z = \lambda x + (1-\lambda) y = \begin{bmatrix} \lambda x_1 + (1-\lambda) y_1 \\ \lambda x_2 + (1-\lambda) y_2 \end{bmatrix}$

$\therefore z^T A z = [\lambda x_1 + (1-\lambda) y_1]^2 - [\lambda x_2 + (1-\lambda) y_2]^2$

$= \lambda^2 (x_1^2 - x_2^2) + (1-\lambda)^2 (y_1^2 - y_2^2) + 2\lambda(1-\lambda)(x_1 y_1 - x_2 y_2)$

A counter example $\quad x_1 = 1 \quad y_1 = 1 \quad x_2 = 1 \quad y_2 = -1$

$x = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$, $y = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$, $\lambda = \frac{1}{2}$

$\therefore z = \frac{1}{2} x + (1 - \frac{1}{2}) y = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$

$z^T A z = 1 > 0$, $\therefore z \notin \{x \in \mathbb{R}^2 : x^T A x \leq 0\}$

$\therefore$ The set is not convex

Problem 2

a) Yes

$\forall \lambda \in [0,1]$

$\because \lambda f(x) + (1-\lambda) f(y) = \sum_{i=1}^{d} \lambda |x_i| + (1-\lambda) |y_i|$

$f(\lambda f(x) + (1-\lambda) f(y)) = \sum_{i=1}^{d} |\lambda x_i + (1-\lambda) y_i|$

$\because \lambda |x_i| + (1-\lambda) |y_i| \geq |\lambda x_i + (1-\lambda) y_i|$

$\therefore f(\lambda f(x) + (1-\lambda) f(y)) \leq \lambda f(x) + (1-\lambda) f(y)$

$\therefore f(x)$ is convex.

b) Yes

$f(x) = \|x - y\|^2 = \sum_{i=1}^{d} (x_i - y_i)^2 = x^T x - 2 y^T x + y^T y$

$\therefore \nabla f(x) = 2x - 2y$

$\nabla^2 f(x) = 2I$

$\therefore \nabla^2 f(x)$ is positive semi-definite

$\therefore f(x)$ is convex.

(c) Yes

$f: \mathbb{R}^1 \to \mathbb{R}$, $f(\lambda x) = |\lambda| f(x)$, $f(x+y) \le f(x) + f(y)$

$\forall \lambda \in [0,1]$

$\therefore f(\lambda x + (1-\lambda) y) \le f(\lambda x) + f((1-\lambda) y)$

$= |\lambda| f(x) + |1-\lambda| f(y)$

$\because \lambda \in [0,1]$, $1-\lambda \in [0,1]$

$\therefore |\lambda| = \lambda$, $|1-\lambda| = 1-\lambda$

$\therefore f(\lambda x + (1-\lambda) y) \le \lambda f(x) + (1-\lambda) f(y)$

$\therefore$ The function is convex

(d) Yes

$f(x) = |x|^{\frac{5}{2}} = \begin{cases} x^{\frac{5}{2}}, & x \ge 0 \\ (-x)^{\frac{5}{2}}, & x < 0 \end{cases}$

$\therefore f'(x) = \begin{cases} \frac{5}{2} x^{\frac{3}{2}}, & x \ge 0 \\ -\frac{5}{2}(-x)^{\frac{3}{2}}, & x < 0 \end{cases}$

$\therefore f''(x) = \begin{cases} \frac{15}{4} x^{\frac{1}{2}}, & x \ge 0 \\ \frac{15}{4}(-x)^{\frac{1}{2}}, & x < 0 \end{cases}$

$\therefore f''(x) > 0$

$\therefore$ The function is convex.

(e) Yes

$f(x) = \ln\left(x + \sqrt{1+x^2}\right) + x^2 = \ln\left(\sqrt{x^2+1} + \sqrt{x^2}\right) + x^2 = \ln\left(\frac{1}{\sqrt{x^2+1} - \sqrt{x^2}}\right) + x^2$

$\therefore f'(x) = \frac{1 + \frac{x}{\sqrt{1+x^2}}}{x + \sqrt{1+x^2}} + 2x$

$= \frac{\sqrt{1+x^2} + x}{\sqrt{1+x^2}\left(x + \sqrt{1+x^2}\right)} + 2x = \frac{1}{\sqrt{1+x^2}} + 2x$

$\therefore f''(x) = -x(1+x^2)^{-\frac{3}{2}} + 2$

$\therefore$ When $x < 0$, $-x(1+x^2)^{-\frac{3}{2}} > 0$, $f''(x) > 0$

When $x \ge 0$, $2 - \frac{x}{(1+x^2)^{\frac{3}{2}}}$

$= \frac{2(1+x^2)^{\frac{3}{2}} - x}{(1+x^2)^{\frac{3}{2}}} > \frac{2(1+x^2) - x}{(1+x^2)^{\frac{3}{2}}} = \frac{2x^2 - x + 2}{(1+x^2)^{\frac{3}{2}}} = \frac{(\sqrt{2}x - \frac{1}{2\sqrt{2}})^2 + \frac{15}{8}}{(1+x^2)^{\frac{3}{2}}} > 0$

$\therefore f''(x) > 0$ for $x \in \mathbb{R}$

$\therefore f(x)$ is convex

Problem 3

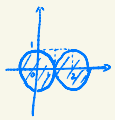
(a) False.

A counter example:

$S_1 = \{(x,y) : x^2 + y^2 = 1\}$, $S_2 = \{(x,y) : (x-2)^2 + y^2 = 1\}$

$(0,1) \in S_1$, $(2,1) \in S_2$

$\therefore \frac{1}{2}[(0,1) + (2,1)] = (1,1) \notin S_1 \cup S_2$



(b) False.

A counter example:

$f(x) = -x$, $g(x) = x^2$

$\therefore h(x) = fg = f(g(x)) = -x^2$

$\therefore h''(x) = -2 < 0$

$\therefore h(x)$ is concave

$\therefore$ False.

(c) False

A counter example:

$f(x) = -x^{\frac{5}{3}}$

$\therefore f'(x) = -\frac{5}{3} x^{\frac{2}{3}}$, $f''(x) = \frac{10}{9} x^{-\frac{1}{3}} > 0$

$g(x) = x^2$

$\therefore f'(x) = 2x$, $f''(x) = 2 > 0$

$\therefore h(x) = f(x) \cdot g(x) = -x^{\frac{11}{3}}$

$\therefore h'(x) = -\frac{11}{3} x^{\frac{8}{3}}$, $h''(x) = -\frac{440}{9} x^{\frac{5}{3}} < 0$

$\therefore h(x)$ may not be convex

(d) False.

A counter example:

$f(x) = x^2$ $\therefore f''(x) = 2$, $f(x)$ is convex

$\therefore (1,1) \in S$, $(4,2) \in S$

$\therefore$ For $\lambda = \frac{1}{2}$, $\frac{1}{2} \cdot (1,1) + \frac{1}{2} \cdot (4,2) = (\frac{5}{2}, \frac{3}{2})$

$\because (\frac{5}{2}, \frac{3}{2}) \notin S$

$\therefore S$ is not a convex set.

e) True.

A Line segment in $\mathbb{R}^2$ can be written as $L = \{\lambda x + (1-\lambda) y \mid \lambda \in [0,1]\}$, with x, y as vertices

$\therefore$ For any $p, q \in L$, $\alpha p + (1-\alpha) q \in L$, $\alpha \in [0,1]$

$\therefore$ A line segment in $\mathbb{R}^2$ is always a convex set.

Problem 4.

a) $f(\beta_0, \beta) = \sum_{i=1}^{n} (\beta_0 + \beta^T x_i - y_i)^2$

$\therefore \frac{\partial f}{\partial \beta_0} = \sum_{i=1}^{n} 2(\beta_0 + \beta^T x_i - y_i)$

$\because \sum_{i=1}^{n} x_i = \sum_{i=1}^{n} y_i = 0$

$\therefore \frac{\partial f}{\partial \beta_0} = 2n\beta_0$

$\because (\beta_0^*, \beta^*)$ is a global minimizer. $\frac{\partial f}{\partial \beta_0^*} = 2n\beta_0^* = 0$

$\therefore \beta_0^* = 0$

b) $\because x_i' = x_i + q$, $y_i' = y_i + r$

$\therefore \tilde{f}(\beta_0, \beta) = \sum_{i=1}^{n} (\beta_0 + \beta^T x_i' - y_i')^2$

$= \sum_{i=1}^{n} (\beta_0 + \beta^T x_i + \beta^T q - y_i - r)$

$= \sum_{i=1}^{n} ((\beta_0 + \beta^T q - r) + \beta^T x_i - y_i)^2$

$\therefore$ Assuming $\tilde{\beta}_0 = \beta_0 + \beta^T q - r$

$\tilde{f}(\beta_0, \beta) = \sum_{i=1}^{n} (\tilde{\beta}_0 + \beta^T x_i - y_i)^2 = f(\tilde{\beta}_0, \beta)$

$\therefore (\beta_0, \beta)$ minimizes $\tilde{f} \iff (\tilde{\beta}_0, \beta)$ minimizes $f \iff (\beta_0 + \beta^T q - r, \beta)$ minimizes f

$\therefore$ When we replace β_0 by $\beta_0 - \beta^T q + r$

Now $\beta_0 \to \beta_0 - \beta^T q + r$, $\tilde{\beta}_0 \to \beta_0 - \beta^T q + r + \beta^T q - r = \beta_0$

$(\beta_0 - \beta^T q + r, \beta)$ minimizes $\tilde{f} \iff (\beta_0, \beta)$ minimizes f.

Problem 5.

$\min_{x_1, x_2} x_1^2 + x_2^2$

subject to $x_1 + 2x_2 \le -3$

a) $L(x, \lambda) = x_1^2 + x_2^2 + \lambda(x_1 + 2x_2 + 3)$

$\therefore \nabla_x L(x, \lambda) = \begin{bmatrix} 2x_1 + \lambda \\ 2x_2 + 2\lambda \end{bmatrix}$

$\therefore$ The main condition:

$\begin{bmatrix} 2x_1 + \lambda \\ 2x_2 + 2\lambda \end{bmatrix} = 0$

Dual feasibility:

$\lambda \ge 0$

Complementarity:

$\lambda(x_1 + 2x_2 + 3) = 0$

Primal feasibility:

$x_1 + 2x_2 \le -3$

$\therefore$ ① $\lambda = 0$

$\therefore x_1 = x_2 = 0$

But $x_1 + 2x_2 = 0 > -3$ $\therefore$ Infeasible

② $\lambda \neq 0$

$\therefore x_1 = -\frac{1}{2}\lambda$, $x_2 = -\lambda$

$x_1 + 2x_2 = -\frac{5}{2}\lambda = -3$

$\therefore \begin{cases} \lambda = \frac{6}{5} \\ x_1 = -\frac{3}{5} \\ x_2 = -\frac{6}{5} \end{cases}$

$\therefore$ The KKT-point: $x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} -\frac{3}{5} \\ -\frac{6}{5} \end{bmatrix}$

b) $\because f(x) = x_1^2 + x_2^2$

$\therefore \nabla f(x) = \begin{bmatrix} 2x_1 \\ 2x_2 \end{bmatrix}$

$\nabla^2 f(x) = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$, which is positive semi-definite.

$\therefore f(x)$ is convex

$\because$ The constraint is linear (convex)

$\because \nabla g(x) = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ is always linearly independent.

$\therefore$ CQ holds

$\therefore$ The KKT point $(-\frac{3}{5}, -\frac{6}{5})$ is the global minimizer.

c) $\because \nabla^2 f(x) = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$

$\therefore f(x)$ is strictly convex

And the feasible set is convex

$\therefore$ The global minimizer is unique.

$\therefore$ There are no other global minimizers.